Creative Design Project II (BECS 31811)

Project Title	Customized Queue Management System
Group	A
Group Members	EC/2022/003-W.L.D.T Induwara EC/2022/024-A.B.A. Shanendri EC/2022/022- NN Rubasinghe EC/2022/005-K.K.R.Deshan EC/2022/018-D.D.Dalugoda EC/2022/002-R.H.M.P.Rathmalage EC/2022/006-A. W.B.S.Kumarage EC/2022/010-K.K.N.Dilshan EC/2022/026-D.K.T.Uthpala

1.Identified Key Limitations

- Monolithic Architecture: Hard-coded for specific industries, requiring extensive rewrites to adapt.
- Rigid Data: Fixed databases unable to handle diverse fields like Account Numbers vs. Patient IDs.
- Static Hardware:Location-bound ESP32 displays requiring manual reprogramming to repurpose.

2.Planned Improvements (Based on Required Categories)

2.1 System Design/Implementation Update

- Multi Tenant SaaS: A shared platform for multiple businesses.
- Flexible Schema:Dynamic NoSQL storage for varied customer data fields.
- Virtual Queuing: Mobile-first ticketing to eliminate physical lines and paper.
- Dynamic UI: Automated white-labeling that adapts to tenant branding.
- Universal Hardware: Plug-and-play ESP32 displays with cloud-synced firmware.
- Integrate an AI module (using NLP like OpenAI or Dialogflow) that can handle multi-lingual support for diverse South Asian users.

2.2 Experimental Testing (Minimum: 2 Test Cases)

- Test Case 01 (Data Isolation): Verification of multi tenant security to ensure Business A cannot access or view any data belonging to Business B.
- Test Case 02 (Hardware Concurrency): Load testing the server's ability to handle simultaneous requests from50+ESP32units to ensure stability and zero downtime.
- Chatbot Accuracy & Context Isolation" test to ensure the AI only provides information relevant to the specific tenant (e.g., a hospital bot shouldn't answer banking questions)

2.3 Analytical Evaluation (Minimum: 2 Analyses)

- Cost Analysis: Comparing high traditional overhead against lower,scalable SaaS costs.

- Latency Analysis:Measuring ESP32 "Request-to-Display" time, targeting <1s response.

- Conduct a "User Experience (UX) Efficiency Analysis" to measure how much the chatbot reduces manual front-desk inquiries.

2.4 Documentation (Choose at least ONE)

- will provide a detailed System Architecture Diagram illustrating the integration between the web dashboard, cloud server, JSON database,and universal hardware. Additionally, a full API Specification will be documented.

2.5 Additional Improvements (Student Defined)

- Retail (Scan & Go): Mobile item scanning paired with a "Trust Score" algorithm that flags high risk transactions for manual verification without extra hardware.

- Law Enforcement (Privacy & Priority):A specialized mode for police stations that masks sensitive case details on public screens and includes a Priority Override for emergency incidents.

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